Introduction to Machine Learning
Homework 1

Songchen Wang Student ID: 2024201594

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Question 1

Answer

We are given N labeled samples $\{(x_n, y_n)\}_{n=1}^N$ with $x_n \in \mathbb{R}$ and $y_n \in \mathbb{R}$. The D-order polynomial regression model is

$$y_n = \sum_{d=1}^{D} x_n^{d-1}\,\theta_d + \epsilon_n = \boldsymbol{\phi}(x_n)^\top \boldsymbol{\theta} + \epsilon_n, \quad \boldsymbol{\phi}(x) \triangleq [1, x, \ldots, x^{D-1}]^\top. \tag{1}$$

The noise is data-dependent:

$$\epsilon_n \mid x_n \sim \mathcal{N}(0,\, x_n^2), \quad p(\epsilon_n \mid x_n) = \frac{1}{\sqrt{2\pi x_n^2}} \exp\left(-\frac{\epsilon_n^2}{2x_n^2}\right). \tag{2}$$

Assume $x_n \neq 0$ (otherwise $\mathrm{Var}(\epsilon_n \mid x_n) = 0$ makes the likelihood degenerate).

Likelihood. Conditioned on x_n and $\boldsymbol{\theta}$,

$$y_n \mid x_n, \boldsymbol{\theta} \sim \mathcal{N}(\boldsymbol{\phi}(x_n)^\top \boldsymbol{\theta},\, x_n^2), \tag{3}$$

hence

$$\log p(\boldsymbol{y} \mid \boldsymbol{x}, \boldsymbol{\theta}) = \sum_{n=1}^{N} \log \mathcal{N}\big(y_n;\, \boldsymbol{\phi}(x_n)^\top \boldsymbol{\theta},\, x_n^2\big) \tag{4}$$

$$= -\frac{1}{2} \sum_{n=1}^{N} \left[\log(2\pi x_n^2) + \frac{\big(y_n - \boldsymbol{\phi}(x_n)^\top \boldsymbol{\theta}\big)^2}{x_n^2}\right]. \tag{5}$$

The MLE maximizes the above, equivalently minimizes

$$\min_{\boldsymbol{\theta}} \sum_{n=1}^{N} \frac{\big(y_n - \boldsymbol{\phi}(x_n)^\top \boldsymbol{\theta}\big)^2}{x_n^2}. \tag{6}$$

Matrix form and closed-form solution. Let $\mathbf{\Phi} \in \mathbb{R}^{N\times D}$ be the design matrix with $\mathbf{\Phi}_{n,:} = \boldsymbol{\phi}(x_n)^\top$, let $\boldsymbol{y} = [y_1, \dots, y_N]^\top$, and define the diagonal weight matrix

$$\boldsymbol{W} \triangleq \operatorname{diag}\left(\frac{1}{x_1^2}, \dots, \frac{1}{x_N^2}\right). \tag{7}$$

Then (6) is the weighted least-squares problem

$$\min_{\boldsymbol{\theta}} \, (\boldsymbol{y} - \mathbf{\Phi}\boldsymbol{\theta})^\top \boldsymbol{W} (\boldsymbol{y} - \mathbf{\Phi}\boldsymbol{\theta}). \tag{8}$$

Taking derivative and setting to zero gives the normal equation

$$\mathbf{\Phi}^\top \boldsymbol{W} \mathbf{\Phi} \, \widehat{\boldsymbol{\theta}}_{\text{MLE}} = \mathbf{\Phi}^\top \boldsymbol{W} \boldsymbol{y}. \tag{9}$$

So the closed-form MLE is

$$\boxed{\widehat{\boldsymbol{\theta}}_{\text{MLE}} = (\mathbf{\Phi}^\top \boldsymbol{W} \mathbf{\Phi})^{-1} \mathbf{\Phi}^\top \boldsymbol{W} \boldsymbol{y}}. \tag{10}$$

Question 2

Answer

Assume independent priors on the coefficients:

$$\theta_d \sim \text{Exp}(b_d), \qquad p(\theta_d) = \frac{1}{b_d} \exp\left(-\frac{\theta_d}{b_d}\right), \quad \theta_d \geq 0, \quad d = 1, \dots, D. \tag{11}$$

Thus

$$\log p(\boldsymbol{\theta}) = \sum_{d=1}^{D} \left(-\log b_d - \frac{\theta_d}{b_d}\right) (\boldsymbol{\theta} \succeq \mathbf{0}), \tag{12}$$

MAP. The MAP estimate maximizes $\log p(\boldsymbol{y} \mid \boldsymbol{x}, \boldsymbol{\theta}) + \log p(\boldsymbol{\theta})$. Dropping constants w.r.t. $\boldsymbol{\theta}$, this is equivalent to

$$\boxed{\widehat{\boldsymbol{\theta}}_{\text{MAP}} = \arg\min_{\boldsymbol{\theta} \succeq \mathbf{0}} \left\{ \frac{1}{2} (\boldsymbol{y} - \mathbf{\Phi}\boldsymbol{\theta})^\top \boldsymbol{W} (\boldsymbol{y} - \mathbf{\Phi}\boldsymbol{\theta}) + \sum_{d=1}^{D} \frac{\theta_d}{b_d} \right\}.} \tag{13}$$

This is a convex quadratic program: a weighted least-squares data term plus a linear penalty. Note that since $\theta_d \geq 0$, we have $\sum_{d=1}^{D} \theta_d / b_d = \sum_{d=1}^{D} (1/b_d)|\theta_d|$, i.e., a weighted ℓ_1 regularizer restricted to the nonnegative orthant.

Learning process: soft-thresholding. Let $\lambda_d \triangleq 1/b_d$. Then (13) becomes

$$\min_{\boldsymbol{\theta} \succeq \mathbf{0}} \, \frac{1}{2} \|\boldsymbol{W}^{1/2} (\boldsymbol{y} - \mathbf{\Phi}\boldsymbol{\theta})\|_2^2 + \sum_{d=1}^{D} \lambda_d \theta_d, \tag{14}$$

which is a weighted LASSO restricted to $\boldsymbol{\theta} \succeq \mathbf{0}$ (since $\theta_d \geq 0 \Rightarrow \lambda_d \theta_d = \lambda_d |\theta_d|$).

Define the one-sided soft-thresholding operator $S_+(z, \tau) \triangleq \max(0, \ z - \tau)$. Denote by $\boldsymbol{\phi}_d$ the d-th column of $\mathbf{\Phi}$ and $a_d \triangleq \boldsymbol{\phi}_d^\top \boldsymbol{W} \boldsymbol{\phi}_d$. Starting from $\boldsymbol{\theta}^{(0)} \succeq \mathbf{0}$, coordinate descent updates one coordinate at a time. Let the current residual be $\boldsymbol{r}^{(t)} \triangleq \boldsymbol{y} - \boldsymbol{\Phi}\boldsymbol{\theta}^{(t)}$. For $d = 1, \ldots, D$, the core iteration is the single equality

$$\boxed{\theta_d^{(t+1)} \;\leftarrow\; S_+\!\left(\theta_d^{(t)} + \frac{\boldsymbol{\phi}_d^\top \boldsymbol{W}\boldsymbol{r}^{(t)}}{a_d},\; \frac{\lambda_d}{a_d}\right) \;=\; \max\!\left(0,\; \theta_d^{(t)} + \frac{\boldsymbol{\phi}_d^\top \boldsymbol{W}\boldsymbol{r}^{(t)}}{a_d} - \frac{\lambda_d}{a_d}\right).} \tag{15}$$

Repeating sweeps until convergence yields $\widehat{\boldsymbol{\theta}}_{\mathrm{MAP}}$.